COMET: CERTIFIED COUNTERFACTUAL AUDITING UNDER WEAK OVERLAP AND BLOCK DEPENDENCE

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ABSTRACT

Rare exposures and correlated observations can make counterfactual point estimates too uncertain for subgroup auditing or constraint certification. We study simultaneous inference for a finite family of counterfactual moments with known exposure propensities, independent evaluation macro-blocks, and sufficient within-block dependency graphs. Policies, auditors and nuisance predictions are fixed before evaluation or learned from an independent pilot. We give two separate certificates: a median-of-means construction with a known conditional finite-p moment envelope and exact-binomial group selection, and a graph-aware mean certificate for bounded outcomes. Both have explicit finite-sample constants and applicability conditions. An expected-risk lower bound applies to an exact clique-replication subclass; a simultaneous coverage event also yields local feasibility and a robust-margin utility comparison for pessimistic policy selection. In a frozen six-cell study with 1,000 independent logs per cell, every eligible procedure covers all families, yet heavy-tail MOM intervals remain vacuous and combined weak overlap and dependence can force complete decision abstention. The analysis distinguishes validity from informativeness. General adaptive feedback, mutually trained nuisance estimators and future deployment trajectories remain outside the proved scope.

1 INTRODUCTION

Counterfactual auditing asks whether a policy's value or a subgroup's outcome would satisfy a declared requirement under a target exposure distribution. A point estimate cannot answer this question when rare logged exposures or correlated records make its uncertainty large. The relevant distinction is between a valid certificate and a certificate narrow enough to support a decision.

We study a finite family of counterfactual moments under a fixed exogenous evaluation design. Exposure propensities are known, evaluation macro-blocks are independent, and a sufficient dependency graph describes dependence within each block. Policies, auditors and outcome predictions are fixed before evaluation; an independent pilot may supply them. This model permits weak positive overlap and finite-p score moments while keeping the conditioning needed for finite-sample inference explicit.

COMET provides two separate inference routes. A graph-adjusted median of means (MOM) uses a known conditional moment envelope and a common target across blocks. Its number of groups can be chosen by an exact binomial majority bound. For bounded outcomes, a graph-aware ordinary mean instead uses a known variance upper bound and Chebyshev's inequality. Each procedure spends its own confidence budget over the whole frozen family. Neither choosing the narrower observed interval nor pooling their guarantees is justified without an additional argument.

The contribution is an explicit proof-to-computation contract for these counterfactual certificates: the target, dependence assumptions, constants, confidence budget, feasible group counts, and refusal conditions are stated together. The underlying DR identity, MOM argument, graph coloring and pessimistic selection are established tools, not new concentration principles. An exact-replication subclass supplies an expected-risk lower bound with an explicit constant; it does not establish graph-specific optimality. On a simultaneous coverage event, the same moment interface supports held-out

subgroup tests and local constrained selection. Empirical discovery and repair remain separate future studies.

The frozen six-cell study evaluates coverage, width, non-vacuity and abstention under the declared model. It does not simulate feedback-adaptive histories, general network interference or learned cross-fitting. The result section reports the complete independently validated confirmatory outputs; integration preflight observations are excluded from its coverage denominators.

2 Counterfactual Moments under an Exogenous Block Design

Let $\mathcal{D}$ contain a design fixed before evaluation randomization and, optionally, pilot data independent of the evaluation experiment conditional on that design. All statements below condition on $\mathcal{D}$. It fixes the retained units, contexts, references $f_u \in [0,1]$, a finite family Θ of m pairs $\theta = (q,h)$, and predictions $\widehat{\mu}_u(s) \in [0,1]$. We allow $|h_u| \le 1$; reported subgroup moments use binary auditors. No evaluation outcome or future outcome-derived context enters $\mathcal{D}$.

For a finite exposure space $\mathcal{S}$, let $Y_u(s)$ denote the potential outcome and $Y_u = Y_u(S_u)$. The known logging probability is $e_u(s) = \mathbb{P}(S_u = s \mid \mathcal{D})$, with $q_u(s) \ge 0$, $\sum_s q_u(s) = 1$, and $q_u(s) > 0 \Rightarrow e_u(s) > 0$. Assume $Y_u(s) \perp S_u \mid \mathcal{D}$. Write $\mu_u(s) = \mathbb{E}[Y_u(s) \mid \mathcal{D}]$. Our target is

$$M_{\mathcal{D}}(q,h;f) = \frac{1}{N}\sum_u h_u \left[\sum_s q_u(s)\mu_u(s) - f_u\right]. \tag{1}$$

This is a conditional finite-design mean, not a realized-potential-outcome average or the value along an arbitrary adaptive deployment path.

Exposure probabilities and implementable policies. If $S_u = g_u(A)$ arises from a joint assignment, then $e_u(s) = \sum_{a:g_u(a)=s} P(A = a \mid \mathcal{D})$. Products of item marginals generally do not equal these probabilities. To call q a globally implementable policy, require a valid joint law $Q(A \mid \mathcal{D})$ inducing every $q_u(s) = \sum_{a:g_u(a)=s} Q(a \mid \mathcal{D})$. Two units competing for one display slot cannot both have target display probability one. Without such a joint construction, (1) is an exposure-marginal functional. The frozen simulator uses the simpler binary exposure $S = A$.

Score and identification. With a fit fixed given $\mathcal{D}$, define

$$Z_u(\theta) = h_u\left[\sum_s q_u(s)\widehat{\mu}_u(s) - f_u + \frac{q_u(S_u)}{e_u(S_u)}\{Y_u - \widehat{\mu}_u(S_u)\}\right]. \tag{2}$$

Conditional randomization gives $\mathbb{E}[Z_u \mid \mathcal{D}] = m_u(\theta) := h_u[\sum_s q_u(s)\mu_u(s) - f_u]$. This exact-propensity DR identity does not require a correct outcome fit. It does require a legitimate fixed/independent-pilot fit for the concentration results below.

Dependence and moment conditions. There are B conditionally independent evaluation-block vectors. A sufficient dependency graph within each block means that disjoint vertex sets with no connecting edges have independent score vectors given $\mathcal{D}$. Pairwise zero covariance or a network adjacency list alone does not establish this property. A proper coloring uses χ_b colors. The MOM route additionally requires equal block size n_0, $\chi_b \le \chi$, stable block means $n_0^{-1}\sum_{u\in b} m_u(\theta) = M_{\mathcal{D}}(\theta)$, and a known finite envelope

$$\mathbb{E}[|Z_u(\theta) - m_u(\theta)|^p \mid \mathcal{D}] \le \nu_\theta^p, \qquad 1 < p \le 2. \tag{3}$$

An unconditional integrated moment does not imply this uniform conditional bound when q/e is unbounded across designs. The envelope may depend on the realized exogenous design and must be reported. The bounded-outcome mean route instead assumes $Y_u(s) \in [0,1]$ and permits unequal block means.

Subgroups. For binary h, the design mass $\pi_{\mathcal{D},h} = N^{-1}\sum_u h_u$ is known. A numerator interval divided by positive $\pi_{\mathcal{D},h}$ certifies the corresponding fixed-design subgroup mean. At zero mass the ratio is undefined. No superpopulation mass theorem is claimed.

3 TWO CERTIFIED PROCEDURES

Both branches use (2), exact e, a frozen finite family, and a permissible fixed or independent-pilot fit. The implemented study uses the predeclared fit $(.5, .5)$; it does not fit a nuisance model on evaluation blocks.

Finite-p MOM. Partition all B complete blocks, before outcomes, into K groups with lengths $L_g \in \{\lfloor B/K \rfloor, \lceil B/K \rceil\}$. Require $B \geq 2K$ and put $L_{\min} = \lfloor B/K \rfloor$. For each candidate, average scores within blocks and block means within each group using its actual length. Let $\widehat{M}_\theta^{\mathrm{MOM}}$ be the median of these group means. The historical rule uses $K = \lceil 8 \log(2m/\alpha) \rceil$ and averages the central two values if K is even. The exact-majority rule uses the smallest feasible odd K satisfying

$$m \sum_{j=(K+1)/2}^{K} \binom{K}{j} (1/8)^j (7/8)^{K-j} \leq \alpha. \tag{4}$$

Both use the explicit half-width

$$r_\theta^{\mathrm{MOM}} = 32^{1/p} \nu_\theta \left(\frac{\chi}{L_{\min} n_0} \right)^{(p-1)/p}. \tag{5}$$

The exact selector uses rational arithmetic for its binomial comparison. Score and radius calculations use ordinary floating-point arithmetic; the implementation is not an interval-arithmetic proof for every rounding bit. Infeasible block counts cause refusal at the requested level, not a changed confidence budget. Every complete block is retained.

Bounded-outcome graph mean. For $Y, \widehat{\mu} \in [0, 1]$, set

$$v_{u,\theta} = h_u^2 \sum_{s: e_u(s) > 0} \frac{q_u(s)^2}{e_u(s)}, \quad V_\theta = \frac{1}{N^2} \sum_b \chi_b \sum_{u \in b} v_{u,\theta}, \quad r_\theta^{\mathrm{GV}} = \sqrt{m V_\theta / \alpha}. \tag{6}$$

The center is the raw-row mean $N^{-1} \sum_u Z_u(\theta)$. This procedure uses no MOM grouping and no empirical variance estimate. It is inapplicable to the unbounded-outcome F model even if one observed sample happens to lie in $[0, 1]$.

Known scales, intervals and decisions. The model supplies p and ν_θ; empirical tail summaries never enter a certified radius. For bounded rewards a valid choice is $\nu_\theta^p = \max_u 2^{2p-1}[1 + \sum_s q_u(s)^p e_u(s)^{1-p}]$. Appendix A gives the heavy-tail envelope. A missing target-supported exposure is nonidentification. An invalid model, unknown moment envelope or infeasible grouping receives an explicit refusal status; IPS, DM and ordinary DR comparators are point-only.

Report the raw confidence interval and, when a deterministic target range is known, its intersection with that range. For binary h and known $0 \leq \mu_u(s) \leq 1$, the range is $[-N^{-1} \sum_u h_u f_u, N^{-1} \sum_u h_u (1 - f_u)]$, even if individual rewards are unbounded. An interval equal to this full range is vacuous; an empty intersection is a coverage failure. For a constraint $M \leq t$, certify safe if the upper endpoint is at most t, certify violation if the lower endpoint exceeds t, and otherwise abstain. Each method and each confidence level has its own within-family guarantee. Data-dependent selection among their intervals is not covered by those separate statements.

4 FINITE-SAMPLE GUARANTEES

Theorem 4.1 (Independent-block MOM certificate)**.** *Under the MOM conditions in Section 2, use the complete-block partition, historical group count and radius in Section 3. If $B \geq 2K$, then*

$$\mathbb{P}\left(\forall \theta \in \Theta : |\widehat{M}_\theta^{\mathrm{MOM}} - M_{\mathcal{D}}(\theta)| \leq r_\theta^{\mathrm{MOM}} \mid \mathcal{D} \right) \geq 1 - \alpha. \tag{7}$$

Theorem 4.2 (Exact-binomial MOM certificate)**.** *Under the same conditions, replace the historical group count by the smallest feasible odd K in (4). The same statement (7) holds, with its own $L_{\min}$ and unchanged explicit constant $32^{1/p}$. If no such K exists, this procedure issues no certificate.*

Proof. Write $X_u = Z_u - m_u$. Within each independent color class, von Bahr–Esseen gives $\mathbb{E}|\sum X_u|^p \le 2\sum \mathbb{E}|X_u|^p$. Convexity across at most χ color sums yields $\mathbb{E}|\bar{Z}_b - M_{\mathcal{D}}|^p \le 2\nu_\theta^p(\chi/n_0)^{p-1}$. Applying the same inequality across the independent blocks in group g gives $\mathbb{E}|T_g - M_{\mathcal{D}}|^p \le 4\nu_\theta^p(\chi/(L_g n_0))^{p-1}$. All expectations are conditional on $\mathcal{D}$. Markov at (5) bounds each group's failure probability by $1/8$; zero envelope gives exact estimation directly. Group failures are independent, though not identically distributed, and are stochastically dominated by independent Bernoulli$(1/8)$ indicators. For odd K a failed median requires at least $(K+1)/2$ bad groups, proving the exact tail. For the historical rule, fewer than $K/2$ failures place both central order statistics inside the target interval; Hoeffding gives failure at most $e^{-9K/32} \le e^{-K/8}$. The historical family bound is $me^{-K/8} \le \alpha/2$, while (4) bounds the exact-rule family failure by α. Neither union bound requires independent candidates. □

Theorem 4.3 (Bounded-outcome graph variance certificate)**.** *Under the bounded-outcome conditions in Section 2, the mean and radius* (6) *satisfy*

$$\mathbb{P}\left(\forall\theta : \left|N^{-1}\sum_u Z_u(\theta) - M_{\mathcal{D}}(\theta)\right| \le r_\theta^{\mathrm{GV}} \mid \mathcal{D}\right) \ge 1-\alpha.$$

Stable block means are unnecessary for this theorem.

Proof. The direct score term is fixed. Since $|Y - \widehat{\mu}| \le 1$, the conditional second moment of the weighted residual bounds $\mathrm{Var}(Z_u \mid \mathcal{D})$ by $v_{u,\theta}$. Within a color class the centered scores are mutually independent. Applying $(\sum_{c=1}^{\chi_b} T_c)^2 \le \chi_b \sum_c T_c^2$ to the color sums gives $\mathrm{Var}(\sum_{u\in b} Z_u \mid \mathcal{D}) \le \chi_b \sum_{u\in b} v_{u,\theta}$. Block independence then bounds the mean variance by V_θ. Chebyshev gives failure at most α/m for each candidate; union bounding proves the claim. If $V_\theta = 0$, the centered mean is zero almost surely. □

All three conditional statements integrate to unconditional coverage of the same random conditional target $M_{\mathcal{D}}$. They do not turn it into an unconditional population target. Coupling failure is zero in this model. A future coupling construction would spend failure probability from α; it cannot be inserted as a numerical half-width without a separate argument.

Theorem 4.4 (Expected-risk lower bound for exact clique replication)**.** *Let $1 < p \le 2$, $\chi \mid N$, and observe $n = N/\chi$ independent scalar draws, each copied exactly χ times. Over distributions with $\mathbb{E}|X - \mathbb{E}X|^p \le \nu_p^p$,*

$$\inf_T \sup_P \mathbb{E}_P|T - \mathbb{E}_P X| \ge \frac{\nu_p}{8}\left(\frac{\chi}{N}\right)^{(p-1)/p}.$$

Appendix B proves the explicit constant. The scalar mean problem embeds in an unbounded counterfactual-moment class with one exposure, $q = e = 1$, $h = 1$ and $f = 0$. The lower bound concerns this subclass and containing classes; it is not a sharp bound for every sparse graph, a lower bound for a specified 95% interval, or a lower bound for the particular bounded-mean-class F simulator at every sample size.

5 Held-Out Auditing and Local Policy Selection

An independent discovery pilot may choose auditors, target policies and bounded nuisance fits. Freeze their definitions before evaluation and include the pilot in $\mathcal{D}$. Conditional evaluation-block independence must still hold. A temporal guard or mutually trained evaluation folds alone does not prove this premise. Certification pays for the full frozen policy–auditor–constraint family; selecting the top L auditors does not remove the policy and constraint multiplicities.

For distinct utility and cost channels, let $V(q)$ and $C_j(q)$ be fixed-design moments over a finite policy library. Suppose a single simultaneous event covers every utility and cost estimate:

$$|\widehat{V}(q) - V(q)| \le r_V(q), \qquad |\widehat{C}_j(q) - C_j(q)| \le r_j(q) \quad \text{for all } q, j. \tag{8}$$

With J costs, a direct application uses $m = |\mathcal{Q}|(J+1)$ coordinates, or an explicitly proved allocation summing to the total confidence budget. Define the certified feasible library by $\widehat{C}_j(q) + r_j(q) \le a_j$

for all j, and maximize $\widehat{V}(q) - r_V(q)$ within it. If it is empty, abstain or use a separately justified safe fallback; the theorem does not create a feasible candidate.

Proposition 5.1 (Eventwise fixed-design feasibility and utility)**.** *On* (8)*, every selected policy satisfies* $C_j(\widehat{q}) \le a_j$ *for all* j*. For every comparator* q *with* $C_j(q) + 2r_j(q) \le a_j$ *for all* j*,* $V(\widehat{q}) \ge V(q) - 2r_V(q)$*.*

Proof. The upper confidence bound exceeds the true cost on the event, proving feasibility. The comparator's margin gives $\widehat{C}_j(q) + r_j(q) \le C_j(q) + 2r_j(q) \le a_j$, so it is in the certified library. Maximization and the two utility bounds give $V(\widehat{q}) \ge \widehat{V}(\widehat{q}) - r_V(\widehat{q}) \ge \widehat{V}(q) - r_V(q) \ge V(q) - 2r_V(q)$. □

This deterministic consequence concerns the same design and candidate library as the certificate. It does not compare automatically to the best truly feasible boundary policy, nor guarantee a future trajectory after deployment changes contexts. Discovery and repair performance require the separately planned D2/D3 experiments; the six-cell C study does not test them.

6 Frozen Confirmatory Evaluation

Design and completeness. The frozen study completed all 1,000 independent logs in each of six cells, with 128 independent macro-blocks per log. A uses $N = 8192$, $\chi = 1$ and one candidate. B–F use $N = 32768$ and sixteen candidates: eight target exposure probabilities crossed with two fixed auditors. B/C/F have logging probabilities $(.4, .6)$ by context, D/E have $(.01, .02)$, and A uses $(.5, .5)$. C/E/F replicate each independent cluster four times; the other cells use $\chi = 1$. All fits are fixed at .5 and references at .25. A–E have Bernoulli rewards. F adds symmetric Pareto noise with exponent 1.75, scale .1 and known moment order $p = 1.5$. Its ideal law has infinite variance; the finite-grid machine law does not literally have infinite representable support. No outcome clipping is used.

All seven methods and both confidence levels share each log, producing 1,134,000 candidate records. The twelve preflight logs are excluded. Immutable-checkpoint resume generated zero new logs and preserved all 6,000 checkpoint files byte-for-byte. An independent aggregation checked all assigned IDs, statuses and scientific records and matched all 84 summary rows. The scientific model, methods, seeds and preregistered analyses were unchanged.

Family coverage and uncertainty. Each of the seventeen prespecified eligible cell–method comparisons at nominal 95% covered its entire family in 1000/1000 logs. Each descriptive Wilson 95% interval is $[0.996173, 1]$. The exact inclusive lower-tail undercoverage p-values and their Holm adjustments across exactly seventeen tests are all one. These conservative results neither prove nominal validity empirically nor imply useful decisions. The independent observation is one whole log family; its sixteen candidate indicators are correlated. Bounded graph Chebyshev is `NOT_APPLICABLE` in F. IPS, DM and ordinary DR remain `POINT_ONLY`, while the full target range is a deterministic mean-class benchmark, not a concentration result.

Informativeness and decisions. Table 1 separates coverage from practical information. Exact-majority MOM reduces width in A/B but makes no certified constraint decision in any cell. The graph mean is narrower in these frozen bounded-outcome comparisons, with strong dependence or weak overlap reducing its usefulness. This is not a universal dominance claim: the graph radius pays $\sqrt{m}$, whereas MOM's grouping pays a logarithmic family-size factor.

At the fixed constraint $M_{\mathcal{D}} \le .2\pi_h$, the 95% graph method detects at least one true violation in 12/1000 A logs (Wilson interval $[.006878, .020857]$), 1000/1000 B logs ($[.996173, 1]$), 3/1000 C logs ($[.001021, .008783]$), and no D or E logs ($[0, .003827]$ each). B's separate candidate-level detection fraction is 70% among truly violating candidates; it is not the probability of detection per independent log. Every eligible procedure has zero observed false-safe or false-violation family events; the Wilson interval for each zero-of-1000 event is $[0, .003827]$, not a zero-risk statement.

E's graph intervals have median normalized width one and abstain on every candidate despite some non-full-range intersections. Both MOM procedures in F return the full feasible range and always

Cell	Procedure	Width/diam.	Nontrivial	Abstain
A–F	Historical MOM	1.000	0.000%	100.000%
A	Exact MOM	0.873	100.000%	100.000%
A	Graph mean	0.099	100.000%	98.800%
B	Exact MOM	0.916	50.000%	100.000%
B	Graph mean	0.269	100.000%	50.994%
C	Exact MOM	1.000	0.000%	100.000%
C	Graph mean	0.531	100.000%	99.981%
D	Exact MOM	1.000	11.188%	100.000%
D	Graph mean	0.866	61.806%	93.794%
E	Exact MOM	1.000	0.000%	100.000%
E	Graph mean	1.000	31.244%	100.000%
F	Exact MOM	1.000	0.000%	100.000%
F	Graph mean (N/A)	–	–	–

Table 1: Nominal 95% results. Width is the median target-range-intersected width divided by the known feasible diameter. Nontriviality and abstention are descriptive candidate-level fractions, not independent-replication rates. Every eligible row has 1000/1000 family coverage; the first row summarizes the same values in all six cells. N/A is retained without fabricated intervals.

abstain; the bounded-outcome method is inapplicable. These unfavorable findings remain part of the primary comparison. Perfect observed coverage alone would conceal them.

Secondary level and resources. At nominal 90%, all seventeen eligible cell–method combinations also have 1000/1000 family coverage; these are secondary summaries, not another jointly corrected seventeen-test confirmation. The graph median normalized widths in A/B/C/D/E are $.0699/.1902/.3803/.6500/1.0000$, with candidate abstention $31.700/37.575/72.513/93.750/100.000\%$. E and F therefore remain decision-limited even at this lower nominal level. The repository retains raw and intersected width quantiles, some/all family nontriviality, marginal coverage, error/radius diagnostics and all statuses.

The six-worker main batch took 10,389.879 wall seconds and 55,394.379 parent-plus-child CPU seconds. The subsequent zero-draw resume took 288.065 wall seconds and 282.354 CPU seconds, for combined totals of 10,677.944 wall seconds and 55,676.733 CPU seconds. Across these two invocations, the maximum parent peak RSS was 211.500 MiB and the maximum individual worker peak was 116.828 MiB; their separate high-water marks do not measure simultaneous total memory. Checkpoints and both invocation resource records are retained. Five regenerated figure families show coverage uncertainty, widths, nontriviality/abstention, error/radius and resource cost.

Planned studies. D1 varies violation margins and group prevalence; D2 tests independent discover–freeze–certify; D3 tests competing utility/cost selection; D4 requires a new proof for shared primitives and compatible joint policies; D5 studies overlap/tail feasibility and independent-pilot nuisance fits. These and large real-data studies remain unrun. No fixed-Pareto error slope is claimed to establish minimax sharpness.

7 Related Work

Off-policy inference. Doubly robust scores combine regression and importance weighting (Dudík et al., 2011; Jiang and Li, 2016). Karampatziakis, Mineiro and Ramdas develop contextual-bandit confidence sequences valid uniformly over time and at arbitrary stopping times (Karampatziakis et al., 2021). Our fixed-window block certificates do not provide that time-uniform guarantee. Mehrabi and Wager study truncated doubly robust evaluation in Markov decision processes under weak distributional overlap, including slower-rate regimes (Mehrabi and Wager, 2024). Their stationary-distribution-ratio setting differs from our design-conditional finite-exposure moments; COMET does not subsume their MDP analysis.

Heavy tails and dependence. Uniform finite-p MOM estimation is itself established: Høgsgaard and Paudice analyze simultaneous means over function classes with $p \in (1, 2]$ (Høgsgaard and

Paudice, 2025). Our finite-family union bound and exact-binomial group selection should be read as an explicit implementation of standard tools under a different block/graph contract. Graph-dependent concentration and robust mean estimation also predate this work (Janson, 2004; Lugosi and Mendelson, 2019). Neither coloring plus Chebyshev nor the two von Bahr–Esseen steps is claimed as a new concentration inequality.

Combinatorial actions and application scope. Shimizu et al. study contextual combinatorial-bandit OPE/learning by factoring actions and separating main and residual effects (Shimizu et al., 2024). Their work motivates careful action-space modeling; it does not make arbitrary local target marginals jointly implementable. COMET's current experiments use binary actions and replicated cliques. A substantive next application contribution would require demonstrated held-out subgroup discovery or constrained selection, or a proved richer dependence model, rather than relabeling the existing components as new algorithms.

8 Discussion and Limitations

The proved result is simultaneous counterfactual moment inference under an exogenous design, independent evaluation blocks, sufficient within-block dependency graphs, exact exposure propensities and fixed or independent-pilot nuisances. Finite-p MOM and bounded-outcome graph variance supply different certificates with explicit constants and applicability conditions. A valid but full-range interval supports abstention, not a useful safety decision.

The analysis does not establish arbitrary adaptive pathwise coverage, mutually trained cross-fitting, unknown tail-scale fitting, hidden cross-block shocks, optimality for every sparse graph or long-run deployment safety. Exposure mappings and joint-policy compatibility require substantive modeling. In the completed frozen experiment, both MOM methods in heavy-tail F remain wholly vacuous, and the bounded-outcome graph method in E always abstains. The broader discovery, repair and network program requires separate proofs and experiments.

Reproducibility Statement

The experiment repository preserves the scientific freeze, configuration, seed manifest and prior Gate A+B+C0/preflight artifacts. The separate C-STAT phase uses the frozen six-cell design and excludes preflight seeds. All 6,000 confirmatory checkpoints and 1,134,000 candidate records were independently validated; a zero-draw resume preserved the checkpoints byte-for-byte. This proposal changes narrative and theorem presentation only; it does not modify the scientific experiment.

Ethics Statement

The framework is intended to diagnose and constrain counterfactual failures in deployed decision systems, but a valid certificate is conditional on the exposure mapping, logged propensities, identification assumptions, auditor family, and data quality. Misspecified interference neighborhoods or unmeasured confounding can create false assurance. Sensitive group labels should be collected and processed only with appropriate legal, organizational, and privacy safeguards. Dataset experiments will follow the licenses and intended-use conditions of the original releases. Policy repair should not be treated as a substitute for domain review, stakeholder consultation, or prospective evaluation.

AI Use Statement

Generative AI tools were used in research ideation and to help reorganize prior technical notes, draft portions of the prose and LaTeX, propose mathematical formulations, and outline theoretical and experimental directions. The authors are responsible for independently checking every definition, proof, citation, implementation, and empirical claim; unverified suggestions must not be presented as established results.

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A KNOWN CONDITIONAL MOMENT ENVELOPES

For bounded rewards, $|Z_u| \le 1 + q_u(S_u)/e_u(S_u)$ and

$$\mathbb{E}[|Z_u|^p \mid \mathcal{D}] \le 2^{p-1} \left[1 + \sum_s q_u(s)^p e_u(s)^{1-p} \right].$$

Convexity and conditional Jensen give the valid centering inequality $\mathbb{E}|Z_u - \mathbb{E}(Z_u \mid \mathcal{D})|^p \le 2^p \mathbb{E}|Z_u|^p$. Centering does not in general reduce a raw pth moment for $1 < p < 2$.

For F, write $Y(s) = \mu(s) + \varepsilon_s$, where $0 \le \mu(s) \le 1$ and $\varepsilon = R\sigma U^{-1/a}$ has independent fair sign R and continuous uniform $U \in (0,1)$. For $p < a$, $k_p = \mathbb{E}|\varepsilon|^p = a\sigma^p/(a-p)$; sign symmetry and $a > 1$ give zero mean. The residual moment is at most $2^{p-1}(1 + k_p)$. Thus a sufficient known centered envelope is

$$\nu_\theta^p = \max_u 2^{2p-1} \left[1 + 2^{p-1}(1 + k_p) \sum_s q_u(s)^p e_u(s)^{1-p} \right].$$

These scales depend on the known law and design, not observed tail estimates or oracle response locations. In the implementation's positive right-endpoint uniform grid, the decreasing-integrand Riemann sum is bounded by the same continuous integral, apart from ordinary arithmetic rounding. The finite computer law is distinguished from the ideal infinite-variance law. The target interval in F uses the known conditional-mean class, not bounded outcomes.

B Clique-Replication Lower Bound

Proof of Theorem 4.4. Put $n = N/\chi$, $\epsilon = 1/(2n)$, $a = \nu_p(2\epsilon)^{-1/p}$, $P_0 = \delta_0$ and $P_1 = (1-\epsilon)\delta_0 + \epsilon\delta_a$. Their mean separation is $\Delta = \epsilon a = (\nu_p/2)n^{-(p-1)/p}$. The nonzero law satisfies

$$\mathbb{E}_1|X - \mathbb{E}_1 X|^p = a^p[(1-\epsilon)\epsilon^p + \epsilon(1-\epsilon)^p] \le 2\epsilon a^p = \nu_p^p,$$

while P_0 has centered moment zero. Since P_0^n is concentrated at zero, $\mathrm{TV}(P_0^n, P_1^n) = 1-(1-\epsilon)^n \le n\epsilon = 1/2$. The common subprobability measure $Q = P_0^n \wedge P_1^n$ has mass at least $1/2$. For any estimator, including its independent randomization,

$$R_0(T) + R_1(T) \ge \int (|T| + |T-\Delta|)\, dQ \ge \Delta/2.$$

Thus $\max_i R_i(T) \ge \Delta/4 = (\nu_p/8)(\chi/N)^{(p-1)/p}$. Exact replication is invertible by choosing one row per cluster, so it preserves the statistical experiment and total variation. The bound holds for every integer $n \ge 1$. □

This is an expected absolute-risk statement for an N-dependent two-point family; a fixed Pareto experiment is not this least-favorable construction.

C Open Extensions

The adaptive-history target $M_{\mathrm{path}} = N^{-1}\sum_u \mathbb{E}[Z_u \mid \mathcal{F}_{t(u)-1}]$ requires a separate concentration argument. Conditioning on the full realized adaptive path or on mutually trained fits does not supply independent blocks. A future coupling proof must first analyze an independent copy law and add its mismatch probability to the total failure budget. No such adaptive, learned-tail or nonzero-coupling certificate is asserted here.